

St.School Of Programs

A 90 days well-Informed training program with Certified Professionals & Exclusive Mentoring sessions.

MERN FULL STACK DETAILED RESOURCE

MERN Stack 2-Month Mastery Roadmap

Phase 1: JavaScript & Web Fundamentals (Week 1)

Goal:

Strong foundation in JavaScript, DOM, and browser concepts.

Best Resources:

- **JavaScript:**

- [JavaScript.info \(Full Course\)](#) — best free structured course.
- [MDN JavaScript Docs](#) — your JS reference bible.

- [YouTube: Namaste JavaScript by Akshay Saini](#) — top-rated visual explainer.

- **Practice:**

- [LeetCode Easy JavaScript](#)
- [Frontend Mentor](#) — for HTML/CSS/JS projects.

Project Ideas:

- Responsive Portfolio Page
- To-do App (LocalStorage)
- Quiz App (HTML + JS)

Phase 2: Node.js + Express.js (Weeks 2–3)

Goal:

Learn backend APIs, routing, authentication, and middleware.

Best Resources:

- **Node.js:**
 - [Node.js Crash Course – Traversy Media](#)
 - [Node.js Official Docs](#)

- [FreeCodeCamp Node.js & Express Full Course](#)

- **Express.js:**

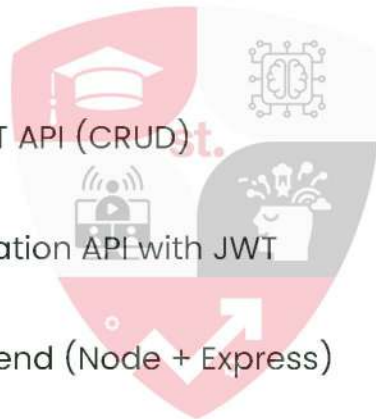
- [Express Official Docs](#)
- [Node + Express + MongoDB REST API – freeCodeCamp Project](#)

- **Authentication:**

- [JWT Auth Tutorial – Dev Ed](#)
- bcrypt and JWT explained

Project Ideas:

- Notes REST API (CRUD)
- Authentication API with JWT
- Blog Backend (Node + Express)



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Phase 3: MongoDB + Mongoose (Week 4)

Goal:

Master NoSQL database, data modeling, and backend integration.

Best Resources:

- [!\[\]\(effbd7993c63c039a58fd3395789cf3f_img.jpg\) MongoDB University Free Course](#)
 - “M001: MongoDB Basics” (official and beginner-friendly)
- [!\[\]\(144980d038f2541d7b588a8a9132bd70_img.jpg\) Mongoose Docs](#)
- [!\[\]\(c4ce2d477989700c971cf3d240ad9283_img.jpg\) MongoDB Crash Course – Traversy Media](#)
- [!\[\]\(5013555a72072875cb154b597e002a46_img.jpg\) Data Modeling Guide \(official\)](#)

Project Ideas:

- Blog Database (Posts + Users)
- E-commerce Products & Users Schema
- Aggregation-based analytics dashboard

Phase 4: React.js Fundamentals (Weeks 5–6)

Goal:

Build single-page UI apps, understand hooks, state, and routing.

Best Resources:

- [!\[\]\(13dd0e1ab3baa23f7c1ed52b3eec2756_img.jpg\) React Official Docs \(New Beta Docs\)](#) — absolute best for fundamentals.

- [!\[\]\(83eb2aa26b610eb6a9dca7cf4702d681_img.jpg\) React Crash Course – Traversy Media](#)
- [!\[\]\(94dfacbf937cdd7da4837a6fcd8fc785_img.jpg\) React Router v6 Docs](#)
- [!\[\]\(dae8c3c5fa7c80febd6526a5e8a853bf_img.jpg\) React Hooks Guide \(freeCodeCamp\)](#)

State Management:

- [Redux Toolkit Docs](#)
- [Redux Simplified Tutorial – Net Ninja](#)

Project Ideas:

- Weather App (API)
- Movie App (TMDB API)
- Todo App (React Hooks + Context)



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Phase 5: Full MERN Integration (Week 7)

Goal:

Connect front-end + back-end into one full-stack app.

Best Resources:

- [!\[\]\(c44db1e92ba1244b2894d325c806ff8a_img.jpg\) MERN Stack Crash Course – Traversy Media \(Best\)](#)

- [!\[\]\(2dc8cdc0c918df88cde61039ecf68682_img.jpg\) MERN Stack Tutorial – freeCodeCamp \(Full Project, 9 hrs\)](#)
- [!\[\]\(793119bf0d613bd9b598fb8668922511_img.jpg\) Net Ninja MERN Course \(Project Based\)](#)
- [!\[\]\(0a4819029e810ca9d2aba79260b63a4d_img.jpg\) JWT Auth MERN Tutorial](#)

Project Ideas:

- Full Blog App
- E-commerce Store
- Chat App (Socket.io + MERN)

Phase 6: Testing, Deployment & Optimization (Week 8)

Goal:

Get production-ready — testing, CI/CD, and performance.

Best Resources:

- **Testing:**
 - [Jest Docs](#)
 - [React Testing Library](#)
- **Deployment:**

- [Deploy MERN App on Vercel + Render](#)
- [MongoDB Atlas Setup](#)
- [CI/CD GitHub Actions Tutorial](#)

- **Optimization:**

- React Performance Guide
- [Node.js Best Practices Repo](#)

Project:

- Deploy your full-stack app live.
- Add testing (Jest + React Testing Library).

Interview Preparation Resources (Top 100 Questions Practice)

Curated Question Practice:

- [InterviewBit Full Stack Developer Questions](#)
- [GitHub – MERN Interview Prep Repository](#)
- [100+ MERN Interview Q&A PDF \(GeeksforGeeks\)](#)

- [🔗 Node.js Interview Questions – Turing.com](#)
- [🔗 React Interview Handbook \(best\)](#)

Books:

- *Full Stack React, Node & MongoDB* by David Katz
 - *Eloquent JavaScript* (for JS logic)
 - *You Don't Know JS* (Kyle Simpson series)
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Daily Practice Tools

- **Code playgrounds:** [Replit](#), [StackBlitz](#)
 - **API Testing:** [Postman](#)
 - **Project Hosting:** [Vercel](#), [Render](#)
 - **Database:** [MongoDB Atlas](#)
 - **Git & Deployment:** [GitHub](#)
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BONUS: MERN Career Prep & Projects Portfolio

Build these before applying for jobs

1. **E-commerce Store** – Full CRUD + payments + JWT Auth
2. **Blog App** – React + Node + MongoDB (Rich Text Editor)
3. **Chat App** – Socket.io + MERN
4. **Student Management System** – Dashboard + Roles + Auth
5. **Task Manager** – React Hooks + JWT

Deploy all projects + write short case studies on GitHub → share links in resume.

PRO ROADMAP IN SHORT:

Phase	Weeks	Key Focus	Best Resource
1	1	JS, HTML, CSS	javascript.info
2	2	Node.js	Traversy Crash Course
3	3	Express.js	freeCodeCamp API Tutorial

4	4	MongoDB	MongoDB University
5	5-6	React.js	react.dev
6	7	MERN Integration	Traversy MERN Project
7	8	Testing + Deployment	Vercel + Jest Guides

Top 100 MERN Stack Interview Questions

Section 1: JavaScript (15 Questions)

1. Difference between `var`, `let`, and `const`.
2. What is a closure in JavaScript?
3. Explain event bubbling and capturing.
4. Difference between synchronous and asynchronous code.
5. What is the event loop in Node.js?

6. Explain promises and async/await.
 7. What are arrow functions and how do they differ from regular functions?
 8. What is hoisting in JavaScript?
 9. What are template literals?
 10. Explain destructuring with an example.
 11. What is a higher-order function?
 12. Explain prototype inheritance.
 13. What are modules in JS (import/export)?
 14. What is debouncing and throttling?
 15. Difference between `null`, `undefined`, and `NaN`.
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Section 2: Node.js (15 Questions)

16. What is Node.js and why is it used?
17. How does Node handle async operations?
18. Explain the Node.js event loop phases.

19. What is a stream in Node.js?
20. How do you manage packages using npm?
21. What are environment variables and how are they used?
22. Difference between `require` and `import`.
23. What are buffers in Node.js?
24. Explain process and child_process modules.
25. How do you handle errors in Node.js?
26. What are REST APIs?
27. What is middleware in Express?
28. How do you create a server in Node.js?
29. How do you secure your Node.js app?
30. What is CORS and how do you enable it?

Section 3: Express.js (15 Questions)

31. What is Express.js?

32. How do you define routes in Express?
33. What is middleware and its types?
34. How to handle 404 errors in Express?
35. What are route parameters?
36. What is the difference between `app.use` and `app.get`?
37. How do you serve static files in Express?
38. How do you implement JWT authentication?
39. What is Helmet and why is it used?
40. How do you connect Express with MongoDB?
41. What are cookies and sessions?
42. How to handle file uploads in Express?
43. How to implement rate limiting?
44. How do you structure large Express apps?
45. Difference between Express and Koa?

Section 4: MongoDB (15 Questions)

46.What is MongoDB and why is it NoSQL?

47.What are collections and documents?

48.Difference between SQL and NoSQL databases.

49.What is Mongoose?

50.How do you perform CRUD operations in MongoDB?

51. What are indexes in MongoDB?

52.What is aggregation?

53.Explain schema design and normalization.

54.What is a replica set?

55.What is sharding?

56.How do you perform data validation in MongoDB?

57.How to populate references in Mongoose?

58.How to handle transactions in MongoDB?

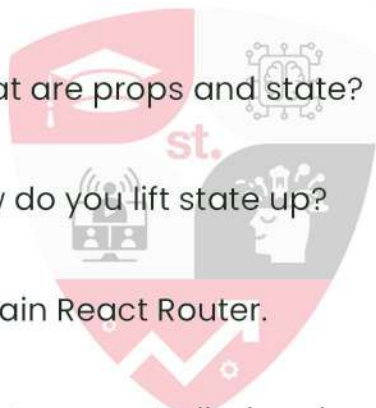
59.What are capped collections?

60.Difference between findOne() and findById().



Section 5: React.js (25 Questions)

- 61. What is React and why use it?
- 62. Difference between class and functional components.
- 63. What are hooks?
- 64. Explain useState and useEffect hooks.
- 65. What is the virtual DOM?
- 66. What is reconciliation in React?
- 67. What are props and state?
- 68. How do you lift state up?
- 69. Explain React Router.
- 70. What are controlled and uncontrolled components?
- 71. What is context API?
- 72. What is Redux and how does it work?
- 73. Explain actions, reducers, and store.
- 74. What is memoization in React?



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75. What are custom hooks?

76. Explain React lifecycle methods.

77. What is lazy loading?

78. What is suspense?

79. What are keys in lists and why are they important?

80. Difference between useMemo and useCallback.

81. How do you optimize React performance?

82. What is React Testing Library?

83. How to handle forms in React?

84. What is a Higher Order Component?

85. What is server-side rendering (SSR)?



Section 6: MERN Integration & Deployment (15 Questions)

86. How do you connect React to Node backend?

87. How do you handle JWT-based authentication in MERN?

- 88.Explain the folder structure of a MERN project.
- 89.How do you protect API routes?
- 90.How do you deploy a MERN app (backend + frontend)?
91. What are environment variables and how to use them in React?
- 92.How do you handle CORS in MERN projects?
- 93.What is proxy in React?
- 94.Explain API rate limiting.
- 95.How do you upload files to the server?
- 96.What are web sockets?
- 97.What is CI/CD and how is it used?
- 98.Explain scaling strategies for MERN apps.
- 99.How to use MongoDB Atlas in production?
100. How do you ensure security in full-stack apps?

Final Tips

- **Build 2–3 real-world projects.** (Blog, E-commerce, Task Manager)
- **Revise core JS daily.** 70% of interview questions revolve around JS logic.
- **Solve 1–2 coding challenges daily** on platforms like LeetCode.
- **Mock interviews:** Pramp, InterviewBit, or peer sessions.
- **Deploy everything.** Employers love deployed, demo-ready projects.



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